Lesson notes · for you

Joining tables

Data engineering · Level 3 · Lesson 3

Time
15 min

Materials
Your phone with Darsi, for the SQL steps; Two sets of small cards: names with numbers, books with numbers

Learning goals

- Join tables with JOIN … ON.
- Keep every row with LEFT JOIN, and find gaps with IS NULL.
- Judge a result: rows missing or counted twice?

Activity: Join with cards

1. Deal (3 min): Half the class gets member cards, half gets loan cards.
2. Inner (4 min): Members and loans with the same number stand together. Who's left sitting?
3. Left (4 min): Now every member stands; those with no loan hold up a NULL card.
4. Count (4 min): Count the pairs, then the different members. Why are the numbers different?

Give one member three loan cards and one loan a number nobody has.

Discussion

- Why did some members vanish in the inner join?
- How do you find members who never borrowed?
- Why did COUNT(*) say 9 when there were 5 borrowers?

Check yourself

- Write a LEFT JOIN that finds pupils with no mark.

Try it this week
Explain a join to someone at home with two sets of cards: names with numbers, and books with numbers. Match them up. Who has no book, and who has two?

Worksheet

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Name _____ Class _____ Date _____

1 **Kofi sat the test and a resit, and appears twice after the join. Is that a mistake?**

◯ Yes: delete one row ◯ Yes: the join is broken ◯ No: one row per matching mark

2 **It's 1 October 2026. Who has an overdue book? Show the member's name and the book for every loan that isn't returned (returned = 0) and was due before '2026-10-01'.**

```
SELECT name, book FROM members
JOIN loans
ON loans.member_id = members.id;
```

Write your program:

3 **Match each piece of a join to what it does.**

JOIN (inner)	• • • •	Says how the rows match
LEFT JOIN	• • • •	Keeps only the rows that match
ON	• • • •	Finds the rows that had no match
IS NULL	• • • •	Keeps every row of the first table

4 **Which members have never borrowed a book? Show their names.**

```
SELECT name FROM members
JOIN loans
ON loans.member_id = members.id
WHERE book IS NULL;
```

Write your program:

5 **This should count the members who have borrowed at least one book (5), but it says 9. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  SELECT COUNT(*)
2  FROM members
3  JOIN loans
4  ON loans.member_id = members.id;
```

✎ **Think of two lists in your life that share an id: a class list and a sports team, a contacts list and a group chat. What would a join of them show? What would a left join show?**

Check your answers

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1 **No: one row per matching mark**

He has two marks, so the join gives two rows, one per match.

2

```
SELECT name, book FROM members
JOIN loans
ON loans.member_id = members.id
WHERE returned = 0
AND due < '2026-10-01';
```

The join brings each loan's member name in; WHERE keeps the loans still out whose due date has passed: Grace's Bird Songs and Kofi's Star Map.

3

JOIN (inner) — Keeps only the rows that match

LEFT JOIN — Keeps every row of the first table

ON — Says how the rows match

IS NULL — Finds the rows that had no match

ON gives the matching rule; the kind of join decides what happens to rows with no partner; IS NULL picks those rows out after a left join.

4

```
SELECT name FROM members
LEFT JOIN loans
ON loans.member_id = members.id
WHERE book IS NULL;
```

LEFT JOIN keeps every member and gives the three with no loans a NULL book; WHERE book IS NULL keeps just them: Tunde, Musa and Ade.

5 **Line 1: `SELECT COUNT(DISTINCT members.id)`**

The join makes one row per loan, so members with several loans are counted several times: 9 loans, 5 members.